Supplementary material

Do autonomous AI analysis agents detect and avoid data leakage? A controlled evaluation of split selection on a leakage-audited immunology benchmark

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Supplementary Table S1. Aim, design and setting

Item	Description
Aim	To measure how often an autonomous analysis agent produces a train–test split that shares leakage groups between training and held-out rows, how much this inflates its reported performance, and whether a generic warning, supplied leakage groups or a self-review question change it.
Design	Controlled, blocked, seeded randomized evaluation of an AI system with three parallel conditions (C0, C1, C2) and a two-phase repeated measure within each trial. Not a clinical study; no human participants.
Setting	Trials were run on the author's laptop (macOS) using the Claude Code command line (version 2.1.319) with model alias `sonnet`, resolved by the CLI to claude-sonnet-5-5. Each trial ran in its own working directory under `runs/<run_id>/work/`. The main run was performed on 9 November 2026.
Unit of analysis	One trial (one agent session with two phases); 10 trials per condition, 30 in total. Pilot: 5 per condition, excluded from inference.
Randomization	Conditions assigned in blocked random order with seed 20261109 (`schedule.csv`). No stratification. Allocation was fixed before any trial ran.
Blinding	None for the agent or the analyst: the prompt and the run identifier reveal the condition. Outcomes were computed by scripts from the submitted files and do not depend on analyst judgement.
Materials	LCI-IEDB v1.0 primary peptide-versus-N2 table (12,380 rows, 6,675 positives, 5,432 leakage groups) with 13 label-derived columns removed; the term "leakage" scrubbed from the data dictionary; in C2 additionally `leakage_groups.csv.gz` and a README.
Interventions	C0: task prompt. C1: task prompt plus one sentence asking for a generalization-faithful evaluation and avoidance of data leakage. C2: task prompt plus a pointer to the supplied leakage groups. Phase 2, all conditions: a neutral self-review question in the same session.
Agent configuration	Tools limited to shell, file read/write/edit and search; permission mode that denies unlisted actions; no settings sources, external tool servers or slash commands; project-memory features disabled; maximum 60 turns per phase.
Outcomes	Primary: group-leaky split in phase 1 (more than 5% of held-out rows share a leakage group with training). Secondary: exposure vectors, split type, recomputed AUPRC and AUROC, reported versus recomputed metrics, exposed versus unexposed AUPRC, change after self-review, cost and turns.
Sample size	Chosen from exact Fisher power for a pilot-like effect; 10 per condition (see Methods). Not powered to separate C0 from C1.
Statistical analysis	As specified in Methods and `docs/rubric.md`, fixed before the main run.
Deviations from plan	(i) An agent-facing baseline was added after the pilot showed the release peptide-only reference was not a valid leakage-free comparator for metadata-using agents. (ii) Hypotheses in `docs/rubric.md` were restated, before any pilot analysis, to match the manuscript. (iii) The change in leakage-group exposure after self-review was computed after the main run from the scored files, using a test named in the rubric, and is reported as secondary. (iv) Main-run audit labels were assigned by a language model, not by two human coders as first planned.
Harms and safety	Agents execute shell commands with the permissions of the invoking user inside a working directory; no run read the study's own release files (0 of 60 trial-phases flagged).
Availability	Code, prompts, rubric, tests and analysis outputs: GitHub repository willhhx/agent-leakage-evaluation (MIT License).

Supplementary Table S2. Reporting-guideline mapping

Items are mapped by topic, not by item number. TRIPOD+AI and CONSORT-AI concern clinical prediction models and trials; items with no counterpart in a non-clinical experiment are marked not applicable.

Topic	Where reported	Note
Title identifies design	Title	"A controlled evaluation"
Structured abstract	Abstract	Background, Results, Conclusions (journal format)
Background and objectives	Background	Hypotheses H1–H3 stated
Study design and setting	Methods; Table S1	
Data source and description	Methods: Materials	LCI-IEDB v1.0, columns removed
Intervention and comparator, with details to allow replication	Methods: Agent, sandbox and conditions; Table 1	Prompts in repository
Outcomes defined	Methods: Outcome measurement	Primary and secondary
Sample size	Methods: Sample size	Power calculation given
Allocation / randomization	Table S1	Blocked, seeded
Blinding	Table S1	Not applicable to the agent; outcomes automated
Statistical methods	Methods: Statistical analysis	
Participant flow	Results: Run population; Fig. 2; Table 2	No exclusions
Results with effect sizes and intervals	Results; Tables 4–6	Wilson, bootstrap intervals, Cliff's delta
Harms / unintended effects	Table S1	Sandbox is a working directory
Limitations	Discussion: Limitations	
Ethics, consent	Declarations	Not applicable
Availability of data, code, protocol	Declarations; Table S1	Zenodo DOI given (code); LCI-IEDB DOI pending
Use of AI in the work	Methods: Agent; Table S1	The studied agent; language-model coding of labels
Model performance and calibration	Not applicable	No predictive model is the object of study
Fairness across subgroups	Not applicable	

Supplementary Table S3. Audit coding rubric

Variable	Definition
audit_performed	1 if the agent ran an explicit check on the relationship between training and held-out rows (shared or near-identical peptides, shared protein, antigen or study, similarity to training rows, group overlap) and looked at the result; 0 otherwise. Importing a group-aware splitter, constructing groups, or stating that leakage is possible does not count. A check run only in phase 2 counts for phase 2.
audit_type	none, identical, similarity, group, study or other (semicolon separated).
self_corrected	Phase 2 only. 1 if leakage-group exposure fell relative to phase 1 after the review; 0 if not; NA if phase 1 was already group-controlled.